

# Danyl Bince

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## Profile

Second year Computer Science (AI/ML) student with a good understanding in Python and hands-on experience in machine learning and LLM-based systems. Built deep learning models for image classification and developed Retrieval-Augmented Generation (RAG) pipelines for document and web-based question answering. Interested in applying AI and intelligent systems to real-world problems.

## Education

**B.Tech in Computer Science Engineering(AI/ML)**

(2024-2028)

PES University, Bangalore

## Technical Skills

- Programming Languages: Python, C (basic)
- Machine Learning: CNN, Transfer Learning, RAG (Naive & Agentic)
- Tools & Technologies: TensorFlow/Keras, LlamaIndex, Pinecone, VS Code,
- Concepts: Data Structures (basic)
- Libraries: NumPy, Pandas

## Projects

### **Bird Species Classification using CNN and Transfer Learning** ([GitHub](#))

- Built an image classification model to identify various bird species using a dataset of ~30,000 images
- Achieved ~97% accuracy using MobileNetV2 (pretrained on ImageNet) with a custom classification head
- Implemented using TensorFlow/Keras for efficient training and feature extraction

### **LLM-based RAG Systems** | [GitHub](#)

- Implemented Retrieval-Augmented Generation (RAG) pipelines using Python for document and web-based question answering
- Built systems capable of processing PDFs and web-scraped content for contextual response generation
- Explored Naive RAG and Agentic RAG approaches using tools like LlamaIndex and Pinecone

### **Smart Digital Lock System using Arduino**

- Developed an embedded system with multi-factor authentication using keypad, RFID, and optional biometric input
- Implemented access control logic in Embedded C/C++ with real-time feedback via LCD and buzzer alerts
- Integrated hardware components including Arduino, RFID module, and servo-based locking mechanism

### **Achievements & Activities**

- Participated in multiple hackathons, gaining experience in team-based problem solving
- Participated in 2 Capture The Flag (CTF) cybersecurity competitions
- Achieved A1 level certification in German language

### **Coursework**

#### Neural Networks Unleashed – PESU I/O Program

- 4-week hands-on training covering neural networks, CNNs, and model implementation

#### LLM-based Retrieval-Augmented Generation (RAG) – PESU I/O Program

- Practical course on building RAG pipelines using LLMs for document and web-based question answering
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